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Project Title:	NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL		
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Space Biology Element:	None		
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Comments:			
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Flight Program:	ISS		
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Task Description:	NOTE: Continuation of "NASA Research under ESA-Based Investigations THERMOLAB and ICOPROSOL," grant number NNX10AU19G. Funding is for Dr. Kelton's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted. Main project: Thermophysical and structural properties and solidification behavior of supercooled glass forming liquids as a member of THERMOLAB; the Science Coordinator is Prof. Dr. Hans Fecht, Ulm University, Institute of Micro and Nanomaterials, Albert-Einstein-Allee 47, Ulm Germany. Similar measurements on undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); the Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum fur Luft- und Raumfahrt (DLR), Institut fur Materialphysik im Weltraum, 51170 Köln, Germany. Subprojects: 1. Ground-Based Experimental Program: Synchrotron X-ray studies of liquid structures are made at the Advanced Photon Source using the ESL (electrostatic levitation) chamber at Washington University (WU-BESL), which has been optimized for scattering studies. These are correlated with neutron scattering data taken at the Spallation Neutron source using the Neutron ESL (NESL), with thermophysical property data obtained on ground with WU-BESL and using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) on the International Space Station (ISS). Terrestrial maximum undercooling and induction time to recalescence data are obtained in ESL, where stirring ensures that nucleation is not diffusion controlled. This data will allow an evaluation of diffusion in the upcoming MSL-EML ISS undercooling studies. 2. Thermophysical Property Measurements: ISS. Specific heat, density, surface tension, viscosity, and electrical resistivity measurements are made on supercooled glass-forming liquids and of Ti-Zr-Ni liquids with different Ni concentrations as a function of the temperature using the MSL-EML facility. These are correlated with x-ray synchrotron and elastic neutron diffraction studies of the supercooled liquids and ground-based measurements made using WU-BESL The purpose is to determine the existence and nature of a cooperative cross-over behavior in the shear viscosity at high temperature, which may be associated with chemical and topological ordering and a possible connection with liquid fragility. Measurements of the specific heat and electrical resistivity are also of interest for their correlation with the dynamical crossover. 3. Effect of diffusion on nucleation: The importance of long-range diffusion on nucleation is being evaluated using the MSL-EML facility located on the ISS. The critical items of interest are the maximum undercooling and the induction time to recalescence in isothermal holds at temperatures below the liquidus temperature. Together with the results from the thermophysical property measurements and an MHD convection model results this will allow us to analyze how the maximum undercooling is influenced by fluid flow. These results of the micro-g experiments are compared with the results of ground-based ESL measurements and with predictions from Kelton’s coupled-flux model for nucleation. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: 1. Ground Based Experiment Program. Measurements of the density, surface tension and viscosity, and synchrotron x-ray diffraction measurements of the liquid structure in the supercooled state using WU-BESL. Ground-based measurements of the maximum undercooling and induction time to recalescence in isothermal holds are also made using WU-BESL. These will be used to evaluate the nature of ordering on phase transitions in the supercooled liquid and fragility. 2. Long Duration Microgravity Experiments. Measurements of thermophysical properties, specific heat (modulated measurements), surface tension, viscosity and density, and also electrical resistivity are made as a function of temperature using the MSL-EML on the ISS. The results are coupled with x-ray and elastic neutron diffraction measurements of the liquid structure factor (WP 2) and ground based studies of the density and viscosity to study the role of ordering on novel liquid/liquid phase transitions and to determine the nature of these transitions. Maximum undercooling and time to recalesce measurements as a function of temperature and stirring will be compared with terrestrial ESL measurements to assess the importance of diffusion on nucleation, allowing Kelton's coupled flux model for nucleation to be evaluated experimentally.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science, encapsulated in the Materials Genome Initiative. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. Even containerless based methods, such as electrostatic levitation, cannot be used to accurately investigate these. The ISS experiments that are being carried out under the support of this grant are allowing these data to be obtained.
	Summary of activities during 2020-2021 Three questions motivated the research that was carried out under the support of this grant during the past year: (1) How does stirring influence crystal nucleation in supercooled metallic liquids when the liquid and primary crystallizing phases have a different chemical composition? (2) Does the specific heat of the liquid reflect the dynamical crossover at temperatures near the melting temperature? (3) Do the thermophysical properties at high temperatures in metallic liquids provide clues on their glass formability? To address these questions, experimental measurements have and are continuing to be made under the microgravity environment available on the International Space Station using the ElectroMagnetic Levitation (EML) facility. These are

Task Progress:

complimented by measurements made under terrestrial conditions using the ElectroStatic Levitation (ESL) facility that is in our laboratory at Washington University. The key accomplishments made during the past year are: (1) An analysis of the data from batch 2 modulation experiments made on the ISS to determine the specific heat as a function of temperature for three metallic liquids, Ti39.5Zr39.5Ni21, Zr64Ni36, Cu50Zr50; (2) the first demonstration of the influence of stirring on nucleation, showing that it increases the nucleation rate in cases where the liquid and nucleating crystal phases have a different chemical composition; (3) preparation for future measurements on batch 3 and batch 4 samples on the ISS.

Measurements of the specific heat in batch 2 samples on the ISS – The ISS-EML facility is uniquely equipped to measure the specific heat of high-melting-temperature alloys in a contamination-free, containerless environment, using the technique of modulation calorimetry. In this technique, the sample is levitated and melted using a radio-frequency electromagnetic field induced by two different sets of water-cooled copper coils that position and heat the sample to the desired temperature. The field from the heating coil is then modulated to induce a temperature modulation in the liquid sample. By knowing the power absorbed by the sample and the coupling constants with the coils, it is possible to extract the specific heat from the measured temperature modulation. It is also possible to measure the specific heat using two different techniques. In one method, the sample is melted and solidified at two different levels of applied power and the specific heat is obtained from the differences in cooling rate, using the known coupling constants with the coils. It is also possible to estimate the hemispherical emissivity from this method. Another method involves decreasing the heating power in small steps so that the sample cools and reaches the equilibrium temperature in each step. The temperature for each step is then fit to an exponential decay; the specific heat can be determined from the decay constant. As mentioned earlier, the specific heats were determined by these three methods for the Ti39.5Zr39.5Ni21, Zr64Ni36, and Cu50Zr50 liquids. The key results are outlined below.

Specific heat for the Ti39.5Zr39.5Ni21 liquid – This liquid crystallizes to a metastable icosahedral quasicrystal phase. The strong icosahedral ordering in the liquid, which catalyzes the nucleation of metastable icosahedral quasicrystal phase, prevents this liquid from forming a metallic glass. Interestingly, unlike the glass forming liquids that show an increase in the specific heat on cooling below the liquidus temperature, this liquid shows a decrease below a temperature, TA, that is located near the liquidus temperature and is believed to correspond to the onset of cooperative dynamical behavior in the liquid. Above this temperature, the specific heat is essentially constant. Further the total hemispherical emissivity shows a near saturation for temperatures above TA. Since the electrical resistivity and emissivity track each other, this supports our observation of a near saturation in the electrical resistivity above TA, which was reported in our last annual report. This high temperature saturation is not predicted from any existing theory of the electrical resistivity.

Specific heat for the Ti64Ni36 liquid – This is a poor glass forming alloy and like Ti39.5Zr39.5Ni21 shows a decrease in the specific heat with decreasing temperature. The decrease is small at high temperatures, but starts to dramatically decrease below the crossover temperature, TA. Unlike Ti39.5Zr39.5Ni21, however, the hemispherical emissivity shows no evidence for a saturation at high temperature. Since the electrical resistivity does show a saturation, the different behavior of the emissivity in this alloy liquid is puzzling and requires further investigation. Specific heat for the Cu50Zr50 liquid – A small temperature dependence in the specific heat is observed at high temperatures, but below TA it increases with decreasing temperature, which is opposite to the trend observed in the Ti39.5Zr39.5Ni21 and Ti64Ni36 liquids. Unlike those liquids, Cu50Zr50 can be formed into a bulk metallic glass. The differences between these liquids may reflect fundamental differences in the thermodynamics of the different liquids, which could provide a new method to identify promising candidates for metallic glass formation.

Influence of stirring on nucleation – experiments on the ISS - Nucleation is the first step in the crystallization of a liquid and is, therefore, technologically important because it plays a critical role in the microstructural development materials and thus their physical properties. Nucleation data are typically analyzed using the Classical Theory of Nucleation (CNT), which has been shown to have several fundamental flaws in its thermodynamic formulation. Further, our computer modeling studies have demonstrated that the kinetic model of the CNT is inadequate for nucleation of a crystal phase from a liquid with a different chemical composition. The Coupled-Flux model for nucleation developed by the Principal Investigator (PI) Kelton predicts that in those cases the nucleation rate will decrease with decreasing stirring in the liquid. The test of this prediction has been and remains a key goal of the ISS and terrestrial investigations made under the support of this grant. In both the CNT and the Coupled-Flux model, the nucleation rate is described by the product of a pre-term, A*, and an exponential function of the work of cluster formation, W (the nucleation barrier). For the same value of W, A* should be smaller if coupled-flux nucleation is dominant over CNT.

The test of the Coupled-Flux model focuses on changes in the nucleation rate with stirring. For small levels of stirring, the nucleation rate is controlled by the coupled processes of attachment at the interface of the nucleating cluster, and of long-range diffusion of atoms of the correct type to the nucleating cluster. With large stirring, the need for long-range diffusion is less. The Coupled-Flux model predicts that the nucleation rate should therefore increase with stirring for cases where the nucleating crystal phase has a different composition from the liquid. There should be little effect of stirring on the nucleation rate of crystal phases that have the same chemical composition as the liquid.

Measurements of the maximum undercooling of the liquid before crystallization begins allows an estimate to be made of the nucleation pre-factor and the nucleation barrier. The data for the Vit106 liquid, which forms a crystal phase of a different chemical concentration from that of the liquid, show that the pre-factor increase dramatically with increased levels of stirring, reflecting a change of three orders of magnitude in the nucleation rate. This is consistent with predictions from the Coupled-Flux model for nucleation. Interestingly, there is little change in the nucleation rate with stirring in the Ti39.5Zr39.5Ni21 liquid, which crystallizes to a crystal phase with the same chemical composition as the liquid. However, instead of increasing, the nucleation rate decreases slightly with increased stirring rate. Our earlier studies have shown that nucleation of the icosahedral phase in Ti39.5Zr39.5Ni21 is catalyzed by icosahedral ordering in the supercooled liquid. Since A* (and hence the nucleation rate) decreases with stirring in this liquid, this may mean that stirring disrupts this liquid ordering. If true, this is the first observation of the influence of stirring on crystal nucleation in this way.

These conclusions were made based on only a few undercooling measurements. Many more are needed to carry out an accurate statistical treatment, as we have recently made from ESL studies of several metallic liquids. We are currently making up to 200 undercooling measurements on one of those liquids, Zr80Pt20 in batch 3 on the ISS, under the support of our new NASA grant, 80NSSC21K1649, which was awarded in August 2021. More measurements will be made on our future batch 3 and 4 samples.

	Other ISS-related activities - The experiences gained from our batch 1 and batch 2 experiments are being applied to our upcoming batch 3 and batch 4 experiments. We have observed that sample contamination occurs during gas (Ar and He) processing on the ISS-EML, which is detrimental to the studies. We have, therefore, redesigned our experiments so that all of the nucleation measurements will be completed under vacuum before the samples are exposed to Ar and He. Among all other properties the specific heat measurements are least affected by such contamination. Because of the long hold times necessary for the modulation calorimetry measurements, gas atmospheres mitigate the evaporation problem to a large extent without compromising the measurements significantly. Finally, for increased precision in the electrical resistivity measurements, all future samples will be contained in cups; cages couple to the RF-fields of the EML. Other ISS studies include the following (1) Participation in the Investigator Working Group (IWG)-14 meeting, held virtually on May 18 – 19, 2021. (2) Batch 3 experiments - Quasi-crystalline Undercooled Alloys for Space Investigation (QUASI) is involved in experiments on four Batch 3 alloys, Cu47Zr47Al6 (ID#41), Ti45Zr45Ni10 (ID#50), Zr80Pt20 (ID# 70), and Pt57.5Cu14.7Ni5.3P22.5 (ID#81). Experiments on the Zr80Pt20 and Pt57.5Cu14.7Ni5.3P22.5 samples are currently in progress. Recently, the science protocols and experimental parameters for the Cu47Zr47Al6 and Ti45Zr45Ni10 samples were developed in collaboration with MUSC. Measurements on these two samples will likely start in the summer of 2022. (3) QUASI is the lead group responsible for a Zr50Pd50 sample (ID#97) scheduled to be processed during batch 4 operations on the ISS-EML facility. QUASI is also a co-PI for the processing of three other samples in batch 4: Pd43Cu27Ni10P20 (ID# 82), Pd82Si18 (ID#135), and Ti40Zr27Ni33 (ID#136). A Pd50Zr50 sample was melted and solidified successfully in parabolic flights during September 2020 to satisfy the requirements for the ISS-EML experiments. From the melting cycles in four parabolic flights, the largest undercooling achieved was 276 K (0.15 of Tl=1873 K), which is smaller than the value of 395 K achieved in the ground-based ESL studies at Washington University. This is most likely due to contamination from the He gas used in these experiments. In May 2021, QUASI provided inputs to the Experiment Scientific Requirements (ESR) document for the four samples that will be processed in batch 4. This document is presently under review. QUASI will make the mass loss measurements on the Zr50Pd50 (ID#97) and Ti40Zr27Ni33 (ID#136) samples using our ESL facility at Washington University. The experiments are planned over the next few months and the results will be submitted to MUSC before May 2022. Samples of the above two compositions for the electronic coupling measurements are scheduled to be delivered to MUSC around February 2022.

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